# Junyu Ren

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## Research Interests

**Verified agentic reasoning:** eliminating agent hallucination with domain-specific formal verification.
**Mathematical foundations for AI:** complexity, topology, and mechanistic interpretability of neural networks.
**AI4M, AI4S:** efficient, cost-effective AI methods for non-incremental mathematical and scientific discovery.
**Post-training:** end-to-end RL for multi-agent systems with independent models associated to different agents.

## Education

**University of Chicago** 2024 – Present
*PhD in Computational and Applied Mathematics.*
Advisor: Lek-Heng Lim. GPA: 4.0/4.0.

**University of Oxford** 2023 – 2024
*MMathPhil, Master of Mathematics and Philosophy.* Distinction.
Thesis: *Koopman Operator Theory and Network-informed Dynamic Mode Decomposition.*
Supervisor: Renaud Lambiotte.

**University of Oxford** 2020 – 2023
*BA, Mathematics and Philosophy.* First Class; cohort ranking 1/20.

## Research Experience

**MATS Fellow (MATS 11)** Sept. 2026 – present
*Mentors: Sabrina Shih and Saad Siddiqui, Safe AI Forum (SAIF).*
Agentic incident confidential verification using formal verification and zero-knowledge proofs.

**Researcher** Jun. 2026 – Oct. 2026
Scaling laws in the number of heterogeneous, specialized models post-trained within a model population.

**Research Assistant, University of Chicago** Apr. 2026 – present
*Supervisor: Risi Kondor; collaborator: Su Hyeong Lee.*
Mesa-optimization by rollout-awareness as a risk in recursive self-improvement.
Funding: UK AI Security Institute (AISI).

**Research Assistant, University of Chicago** Jul. 2025 – present
*Supervisor: Lek-Heng Lim.*
Neural network theory and interpretability using topology and geometry.

## Research and Publications

[*] *Equal contribution / co-first authorship; co-first authors listed in alphabetical order.*

### Published and Accepted

- **Manifold-Weighted Neural Networks.**
  **Junyu Ren**, Lek-Heng Lim.
  *NeurIPS 2026 (accepted).*
- **Optimization Dynamics Imprint Semantic Specificity in Contrastive Embedding Norms.**
  Ziwei Su, **Junyu Ren**, Victor Veitch.
  *NeurIPS 2026 (accepted).*
- **Low-dimensional Topology of Deep Neural Networks.**
  **Junyu Ren**, Lek-Heng Lim.
  *ICML 2026.*
- **EG-VAR: A Path Toward Eliminating LLM Hallucination in Empirical Inference via Tool-Attested Kernel Proofs.**
  **Junyu Ren**.
  *Accepted at three ICML 2026 workshops — three parts of one agenda:*
  *FAGEN — proof-checker feedback for agentic workflows.*
  *TAIGR — formal sidecars as technical governance for high-stakes outputs.*
  *PhilML ("From Prompts to Proof Obligations") — world-directed reasoning as auditable formal commitments.*

- **When Do Transformer Components Compose? Validating a Log-Pool Decomposition Criterion.**
  **Junyu Ren**, Su Hyeong Lee, Risi Kondor.
  *ICML 2026 Workshop PhilML.*
- **Routing-Mediated Structural Pseudo-Alignment in Sparse Mixture-of-Experts.**
  **Junyu Ren**, Su Hyeong Lee, Risi Kondor.
  *ICML 2026 Workshop Mech Interp.*
- **MAPPA: Scaling Multiagent Systems with Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan.
  *ICLR 2026 Workshop RSI.*
- **Agent-as-a-Coach: Towards Fully Agentic, Stateful, and Tool-Augmented Process Rewards.**
  Ed Li*, **Junyu Ren***, Cat Yan, Kerem Goksel.
  *ICLR 2026 Workshop MALGAI.*
- **Time-aware Feature Selection: Adaptive Temporal Masking for Stable Sparse Autoencoder Training.**
  Ed Li*, **Junyu Ren***.
  *ICLR 2025 Workshop XAI4Science.*
- **Unlocking Hierarchical Concept Discovery in Language Models Through Geometric Regularization.**
  Ed Li*, **Junyu Ren***.
  *ICLR 2025 Workshop BuildingTrust.*

**Preprints**

- **Pierce–Birkhoff conjecture is false.**
  Zehua Lai, Lek-Heng Lim, **Junyu Ren**.
  *arXiv:2609.10420, Sep. 2026.*
- **Build Your Personalized Research Group: A Multiagent Framework for Continual and Interactive Science Automation.**
  Ed Li*, **Junyu Ren***, Xintian Pan, Cat Yan, Chuanhao Li, Dirk Bergemann, Zhuoran Yang.
  *arXiv:2510.15624, Oct. 2025.*

**Under Review and In Preparation**

- **Training a 2-token attention is NP-hard.**
  **Junyu Ren**, Zehua Lai, Lek-Heng Lim.
  *Manuscript under review, 2026.*
- **Pierce–Birkhoff conjecture in dimension 3.**
  **Junyu Ren**, Zehua Lai, Lek-Heng Lim.
  *Manuscript in preparation, 2026.*
- **Hodge-decomposed Hochschild cohomology of socle quotient rings.**
  **Junyu Ren**, Lek-Heng Lim, Ke Ye.
  *Manuscript in preparation, 2026.*

## Awards and Honors

**McCormick Fellowship, University of Chicago** 2024
**Gibbs Prize for FHS Mathematics and Philosophy, University of Oxford** 2023
**Oriel College Scholarship, University of Oxford** 2020 – 2024

## Talks & Presentations

- **Low-dimensional topology of deep neural networks.**
  AMS 2026 Spring Eastern Sectional Meeting, Special Session, Boston College, Mar. 2026.
- **Hallucination as a Type Error.**
  IDEAL Workshop on Language Models, University of Illinois Chicago, May 2026.

## Professional Services

Reviewer, ICLR 2027 2027
Reviewer, ICML 2026 *(Gold Reviewer Reward)* 2026
Reviewer, ICML 2026 Workshops 2026
Referee, Journal of the ACM (JACM) Aug. 2026 – present
Referee, Communications Physics (Nature Portfolio) 2026
Referee, The Journal of Geometric Analysis Jun. 2026 – present
Referee, Chaos, Solitons & Fractals Jun. 2026 – present
Referee, Linear Algebra and Its Applications May 2025 – present
Referee, SIAM Journal on Computing Oct. 2025 – present
Referee, SIAM Journal on Matrix Analysis and Applications Oct. 2025 – present
Teaching Assistant, University of Chicago Oct. 2025 – present